

WASP-10b

R-1898

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_130718 · created 2026-08-03T13:57:44+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456491.8382 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.199 +/- 0.061
Transit depth (Rp/Rs)^2	3.96 +/- 2.41 [%]
Orbital Inclination (inc)	85.22 +/- 1.88
Airmass coefficient 1 (a1)	1.11 +/- 0.34
Airmass coefficient 2 (a2)	-0.08 +/- 0.11
Scatter in the residuals of the lightcurve fit is	30.39 %
Best Comparison Star	#1 - [128, 222]
Optimal Aperture	3.0
Optimal Annulus	5.51
Transit Duration (day)	0.052 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.199 ± 0.061 vs 0.1592 (deviation 0.65σ)
Duration fit vs published	0.052 d vs 0.0946 d (deviation 1.25σ)
Mid-time O–C	-3.3 min
Depth SNR	3.3
Residual scatter	30.39 %

Light curve

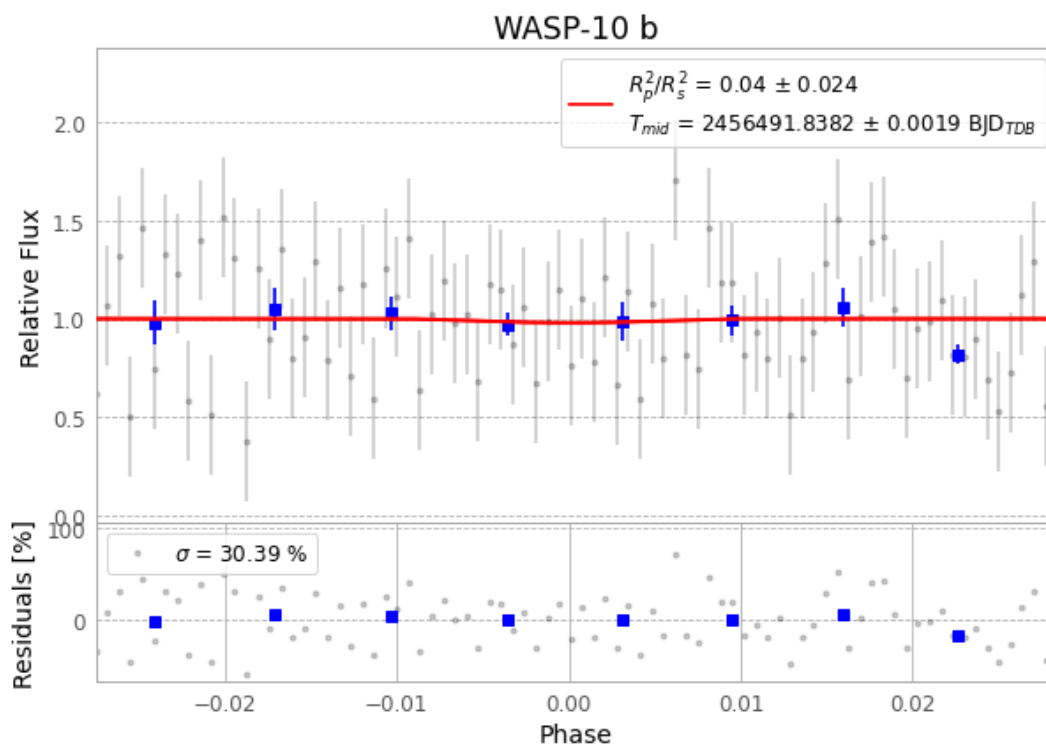


Figure 1 — FinalLightCurve_WASP-10 b_2013-07-17.png (SHA-256 03374e365cd35e38...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2b87cfb964432813...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), WASP-10130718060012.FITS → WASP-10130718100913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-130718.log (b1c34932e444...), FinalLightCurve_WASP-10 b_2013-07-17.png (03374e365cd3...), FinalLightCurve_WASP-10 b_2013-07-17.csv (2902f051191a...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 13:57Z.